

Qatar-1b

R-2462

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110524 · created 2026-08-23T01:11:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455705.85327 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.026
Transit depth (Rp/Rs)^2	1.32 +/- 0.59 [%]
Orbital Inclination (inc)	84.77 +/- 1.93
Airmass coefficient 1 (a1)	1.225 +/- 0.019
Airmass coefficient 2 (a2)	-0.146 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.91
Optimal Annulus	7.72
Transit Duration (day)	0.068 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.026 vs 0.1463 (deviation 1.2 σ)
Duration fit vs published	0.068 d vs 0.0692 d (deviation 0.11 σ)
Mid-time O-C	-1.1 min
Depth SNR	4.4
Residual scatter	1.53 %

Light curve

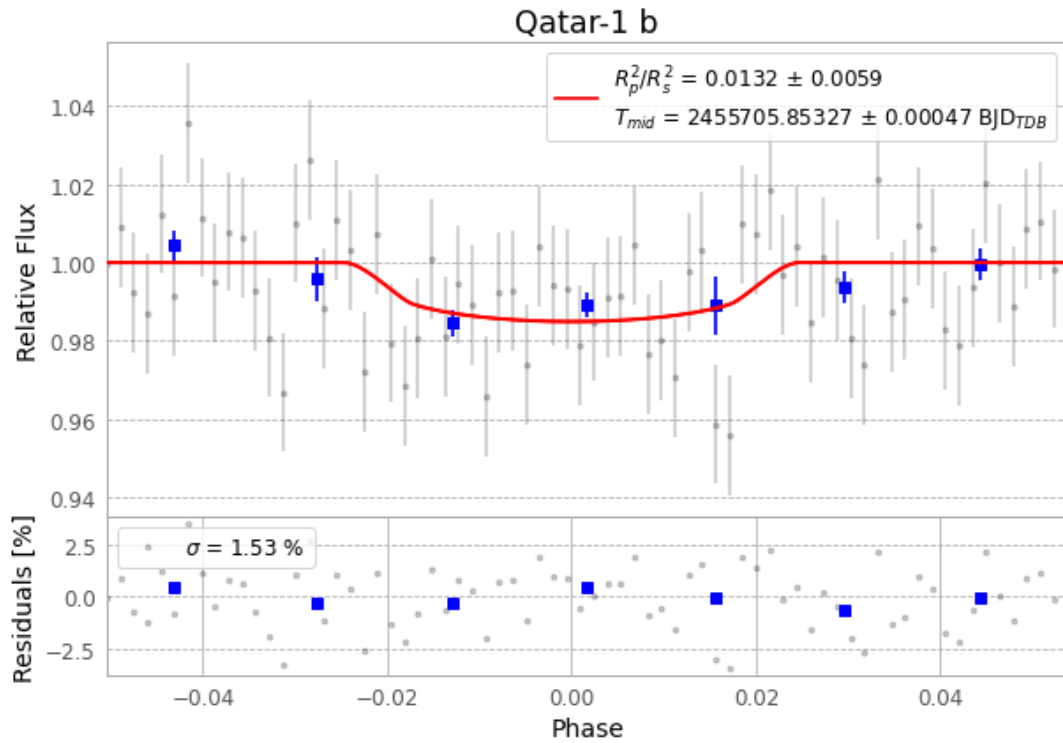


Figure 1 — FinalLightCurve_Qatar-1 b_2011-05-23.png (SHA-256 28e3b219714273b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [444, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c676c9f6307abaea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1110524064513.FITS → Qatar-1110524101813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110524.log (d5337c12411c...), FinalLightCurve_Qatar-1 b_2011-05-23.png (28e3b2197142...), FinalLightCurve_Qatar-1 b_2011-05-23.csv (ea70b3e02d62...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:11Z.